

Data Model and Rating Specification

Master data, entities, premium logic and add-on rules

Project	Property Insurance Placement ERP (working title)
Client	Fiducial (Anila Mary S)
Vendor	Spirezen Enterprises (Hiral Patel)
Document version	v0.2 (Draft)
Date	05 October 2026
Status	Draft for client review

*Version 0.2 is based on the IIB Code Master workbook, the Sookshma and Laghu add-on charts, and the four client workbooks received on 5 Oct 2026 (Data Sheet, RFQ, QCR, Placement Slip). Items still waiting on client answers are marked **[TBC]**. See document 07 for the analysis of the client formats.*

1. Source data summary

Source	Content	Notes
IIB Code sheet	299 occupancies. Columns: S No, TAC code, description, risk grade (RG1-RG9), IIB rate (per mille), Fire risk type, Terrorism risk type, minimum STFI rate, minimum EQ rate for Zone IV, III, II, I.	Codes mix numbers and text (for example 1001_2): store as text. Code 2075 has a blank Terrorism risk type. One row (Multiplex / Malls) is highlighted yellow without explanation.
Pincode sheet	20,611 rows: pincode, state, district, AIFT EQ zone (1-4), EQ rate for Residential, Non-Industrial, Industrial. 43 distinct state values.	5 pincodes duplicated (207001, 302013, 400062, 686654, 742184). Pincode 100000 maps to Gujarat/Kachchh, which looks invalid. Pincode 400001 has state 'Mumbai'.
Add-on charts (images)	15 add-ons for Bharat Sookshma and Bharat Laghu, with cover limit and add-on rate.	Same covers and rate factors; only caps differ.
Client workbooks (5 Oct 2026)	Data Sheet, RFQ, QCR, Placement Slip: 14 coverage sections, annexures, add-on lists (BSUS/BLUS, Fire additional 163, PAR 125, SFSP 185), clauses, excess and warranties.	Product thresholds found in the schedule sheets: BSUS up to 5 Cr, BLUS 5-50 Cr, SFSP above 50 Cr, PAR above 5 Cr. See document 07.

Zone mapping observed: pincode zone 1 corresponds to the 'Zone I' column (highest EQ rate: 0.50 per mille industrial), zone 4 to 'Zone IV' (lowest: 0.05). The pincode sheet's EQ rates by risk type agree with the occupancy sheet's zone columns in the rows checked; the system should verify this for all rows during import.

2. Entity overview

The database is MongoDB. Each entity below is a collection unless noted; lines such as locations, sections, items, annexure items, add-on selections, quote lines and quote conditions are best **embedded** in their parent document (risk or quote), while masters, clients, insurers, rating runs and audit logs are separate collections. Foreign keys are stored as ObjectId references; amounts and rates use Decimal128. See document 04 for the modelling approach.

Entity	Key fields	Notes
organisation	id, name	Single tenant in R1
user, role, user_role	id, email, password_hash, status; role name	RBAC
master_version	id, type, source_file, imported_by, imported_at, status	Groups each master import
occupancy	id, version_id, tac_code (text), description, risk_grade, iib_rate_pm, fire_risk_type, terror_risk_type, min_stfi_pm, min_eq_z4/z3/z2/z1_pm, status, valid_from, valid_to	Status = Preferred / Referred / Declined [TBC]
pincode	id, version_id, pincode, state, district, eq_zone, eq_res_pm, eq_nonind_pm, eq_ind_pm	Unique on (version, pincode)
scheme	id, code (SOOKSHMA / LAGHU), min_si, max_si [TBC]	Selection by total SI
addon_cover	id, scheme_id, seq (1-15), name, description, limit_basis, limit_pct, limit_cap, rate_factor_pct, rate_base_type, rate_base_pct, rate_cap, per_block, version_id	Rules held as data
insurer	id, name, contacts, emails, schemes, active	
client	id, name, group, GST, PAN, address, contacts, owner_user_id	
risk	id, client_id, ref_no, status, policy_period, scheme_id, master_version_ids, created_by	Header record for a placement
location	id, risk_id, address, pincode, state, district, eq_zone, occupancy_id, construction, details	One risk has many locations
sum_insured	id, location_id, class (Building / P&M / F&F / Stocks / Other), amount	
rating_run	id, risk_id, run_at, master_versions, inputs_json, result_json, total_premium	Immutable snapshot

Entity	Key fields	Notes
risk_addon	id, risk_id, addon_id, selected_pct, specified_si, premium	
rfq, rfq_insurer	rfq: id, risk_id, version, sent_at, template_id. rfq_insurer: insurer_id, status, sent_log, due_date	
quote	id, rfq_insurer_id, rates by peril, premium, terms (embedded document), validity, document_id	
qcr	id, rfq_id, recommendation, approved_by, approved_at	
placement_slip	id, slip_no, quote_id, issued_at, status, document_id	Number: FY prefix + sequence
document, template	document: id, type, owner_type, owner_id, storage_key. template: type, version, file	
coverage_section	id, code (FIRE, BURGLARY, FIRE_FLOATER, BURGLARY_FLOATER, FLOP, MONEY, FIDELITY, PLATE_GLASS, NEON, ALL_RISK, EEI, MBD, BOILER, PL), name, line_item_template	14 sections
risk_section	id, risk_id, section_id, option_no (0 = existing, 1, 2), selected, total_si	Sum-insured options per section
section_item	id, risk_section_id, location_id, label, sq_ft, rate_per_sqft, amount	Data Sheet lines (building, FFF, plant and machinery, stocks and so on)
annexure_item	id, risk_section_id, description, make_model, serial_no, year, dimensions, qty, amount	Plate Glass, Neon, All Risk, EEI, MBD, Boiler
product	id, code (BSUS, BLUS, SFSP, PAR), si_min, si_max	Selection thresholds from client schedule
addon_catalogue	id, product_code or section, name, kind (PAID / INBUILT / CLAUSE), limit_text, seq	BSUS/BLUS 15+inbuilt, Fire additional 163, PAR 125, SFSP 185
risk_addon_selection	id, risk_id, catalogue_id, limit_text, selected	
clause	id, section_id, title, wording, default_on	Clauses to be attached
excess_rule	id, product_code, si_from, si_to, pct, min_amt, max_amt, kind	Fire, terrorism, burglary, FLOP excess
existing_policy	id, risk_id, period, insurer, section, sum_insured, premium	Renewal comparison base
claim_history	id, risk_id, period, policy_type, si, premium, claimed_amount, remarks, insurer	Past 3 years
quote_line	id, quote_id, section_id, option_no, sum_insured, premium_with_terror, premium_without_terror	Feeds the QCR matrix
quote_condition	id, quote_id, kind (WARRANTY / EXCESS / OTHER), text	Changes from one insurer to another
placement_share	id, slip_id, insurer_id, capacity_pct, premium	Multiple insurers on one slip [TBC]
audit_log	id, at, user_id, action, entity, entity_id, before, after	Append-only

3. Rating logic (proposed, to be validated with client)

Scope: the IIB-based engine prices the **Fire** section only. The other 13 sections have no rate master in the material received; their premiums are entered from insurer quotes. All rates in the masters are per mille (per thousand of sum insured). For each location, with SI the location's fire sum insured:

Fire premium = SI x IIB rate / 1000
 STFI premium = SI x STFI rate / 1000 (default: occupancy minimum)
 EQ premium = SI x EQ rate / 1000 (pincode rate for the occupancy's Fire risk type)
 Terrorism premium = SI x Terrorism rate / 1000 (rate source TBC)
 Base premium = Fire + STFI + EQ (+ Terrorism if opted)
 Policy rate = Base premium / SI x 1000 (used by add-ons)
 Total before tax = Base premium + Add-on premiums
 GST = Total before tax x GST rate (default 18%, configurable)

Lookup flow: occupancy code gives IIB rate, risk grade, Fire and Terrorism risk types and minimum STFI; pincode gives state, district and EQ zone; the EQ rate is read from the pincode row using the risk type; the system also checks it is not below the occupancy's minimum EQ rate for that zone.

Exceptions: Declined occupancies block quoting; Referred occupancies require manager approval; blank risk types or missing rates are raised as data errors, not silently defaulted.

4. Worked example (illustrative only)

Location in Mumbai (pincode 400001, zone 3), occupancy 2001 Abrasive Manufacturing (IIB 0.69, STFI 0.37, industrial risk type). Total SI Rs 10,00,00,000 (Rs 10 crore, in the BLUS range of 5 to 50 Cr). EQ rate for zone 3, industrial = 0.10.

Component	Working	Amount (Rs)
Fire	$10,00,00,000 \times 0.69 / 1000$	69,000.00
STFI	$10,00,00,000 \times 0.37 / 1000$	37,000.00
Earthquake	$10,00,00,000 \times 0.10 / 1000$	10,000.00
Base premium	Fire + STFI + EQ (policy rate = 1.16 per mille)	1,16,000.00
Add-on 1: Loss minimisation	$1\% \times 1.16 \text{ per mille} \times (5\% \text{ of SI} = 50,00,000)$	58.00
Add-on 6: Escalation (10%)	$50\% \times 1.16 \text{ per mille} \times (10\% \text{ of SI} = 1,00,00,000)$	5,800.00
Total before tax		1,21,858.00
GST at 18%		21,934.44
Total payable		1,43,792.44

Terrorism is excluded because no rate table was supplied. Add-on 1 comes out very small (Rs 58), which is why the add-on reading must be confirmed with the client using their own worked examples before the formulas are built.

5. Add-on cover rules

Limits are shown as Sookshma / Laghu. 'Policy rate' means the combined rate of the base policy [TBC]. 'Rate' is the add-on premium factor from the charts.

#	Add-on	Cover limit (Sookshma / Laghu)	Add-on rate (both schemes)
1	Expenses for loss minimisation / prevention	5% of claim; max 25 L / 1 Cr	1% of policy rate on 5% of SI
2	Hire purchase / lease / consignment property	5% of SI; max 25 L / 1 Cr	2.5% of policy rate on 5% of SI
3	Inadvertent omission	Up to 5% of SI	2.5% of policy rate on 5% of SI
4	Contamination and co-mingling of stocks	Up to 5% of SI	2.5% of policy rate on 5% of SI
5	Expediting expenses	Up to 25 L / 1 Cr	1% of policy rate on 5% of SI (max 25 L / 1 Cr)
6	Escalation (Building, P&M, FFF)	Selected % of SI, max 25% of SI excluding stocks	50% of policy rate on selected % increase
7	Leakage and overflow of oils / chemicals	Up to 1% of stocks SI (AOA) and 10 L / 1 Cr aggregate	5% of policy rate on 1% of SI (max 10 L / 1 Cr)
8	Claims preparation cost	5% of claim; max 5 L / 50 L	5% of policy rate on 5% of SI (max 5 L / 50 L)
9	Involuntary betterment	5% of claim; max 10 L / 1 Cr	5% of policy rate on 5% of SI (max 10 L / 1 Cr)
10	Deterioration of stocks in cold storage	Specified SI; max 25 L / 1 Cr per block	15% of policy rate on specified stocks SI
11	Spoilage material damage cover	Specified SI; max 25 L / 50 L per block	1% of policy rate on specified SI
12	Loss of rent (specified SI)	Specified SI; max 5 L / 10 L for 6 months	1% of policy rate on specified SI

#	Add-on	Cover limit (Sookshma / Laghu)	Add-on rate (both schemes)
13	Immediate repairs	Specified SI; max 5 L / 15 L	1% of policy rate on specified SI
14	Brands and trademark clause	Specified SI; max 10 L / 25 L	1% of policy rate on specified SI
15	Impact damage by insured's own rail/road vehicle	Policy SI (both schemes)	1% of policy rate

The RFQ and QCR workbooks list the same BSUS/BLUS covers with slightly different wording from the charts: expediting is 'max 5% of claim and 25 lakh' versus 'up to 25 lakh', and contamination is 'oil and chemicals only' versus 'stocks'. Four covers are placeholders ('some % on total sum insured'). The client must confirm one version. L = lakh, Cr = crore. Items 1-5, 8, 9 and 15 differ only in the cap between schemes; the rate factors are identical. Cover wording (what each add-on covers) stays as per the client's charts and will be stored as display text.

Calculation types for the engine

Type	Used by	Logic
PCT_OF_SI_BASE	1, 2, 3, 4, 8, 9, 7	Rate factor x policy rate x (base % x SI); cap applied to the cover amount where stated
SPECIFIED_SI	10, 11, 12, 13, 14	Rate factor x policy rate x specified SI; specified SI limited by cap (per block where applicable)
ESCALATION	6	50% x policy rate x (selected % x applicable SI excluding stocks); selected % limited to 25%
POLICY_SI	15	1% x policy rate x policy SI
FLAT_CAP	5	1% x policy rate x 5% of SI, limit the cover to the cap

6. Import and validation rules

- Occupancy: TAC code required and unique per version; rates numeric and not negative; risk grade must be RG1-RG9; risk types must be one of Residential, Industrial, Non-Industrial (blank is flagged).
- Pincode: six digits, unique per version (exact duplicates collapsed with a warning), zone 1-4, rates numeric.
- Cross-check: for each pincode, EQ rate for each risk type equals the occupancy zone columns for the same zone; report mismatches.
- Every import creates a master_version in Draft; an Admin reviews the report and activates it with an effective date.